

Question ID: 3c03cbd8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

A certain township consists of a **5**-hectare industrial park and a **24**-hectare neighborhood. The total number of trees in the township is **4,529**. The equation **$5x + 24y = 4,529$** represents this situation. Which of the following is the best interpretation of **x** in this context?

Answer

- A. The average number of trees per hectare in the industrial park
- B. The average number of trees per hectare in the neighborhood
- C. The total number of trees in the industrial park
- D. The total number of trees in the neighborhood

Correct Answer: A

Rationale

Choice A is correct. It's given that a certain township consists of a **5**-hectare industrial park and a **24**-hectare neighborhood and that the total number of trees in the township is **4,529**. It's also given that the equation **$5x + 24y = 4,529$** represents this situation. Since the total number of trees for a given area can be determined by taking the size of the area, in hectares, times the average number of trees per hectare, the best interpretation of **$5x$** is the number of trees in the industrial park and the best interpretation of **$24y$** is the number of trees in the neighborhood. Since **5** is the size of the industrial park, in hectares, the best interpretation of **x** is the average number of trees per hectare in the industrial park.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.